

NYC TAXI DEMAND FORECASTING



1. Executive Summary & Business Context:

As Gett expands its operations in New York City, optimizing the fleet's deployment to meet fluctuating demand across the city is paramount. The primary objective of this project is to develop a robust machine learning model capable of forecasting taxi demand in every TLC taxi zone at 30-minute intervals.

The core challenge of this project shifts away from standard single-trip predictions to a **time-series data aggregation problem**. We engineered an end-to-end Machine Learning pipeline encompassing data processing, advanced feature engineering, drift detection in a simulated production environment, and an interactive UI dashboard to present forecasts directly to business stakeholders.

2. Data Strategy:

We utilized the open-source NYC TLC "Yellow Taxi" dataset, which required substantial processing to align with Gett's business objectives.

Data Transformation & Aggregation:

The raw dataset consists of individual trip records. Since our objective is to predict overall demand, we fundamentally restructured the problem into a time-series forecasting task. We processed the raw data by aggregating individual rides into **30-minute demand buckets**, for example:

20:02 → 20:00

20:17 → 20:30

20:46 → 21:00

This aggregation converts trip-level data into demand counts per half-hour, enabling demand forecasting at operationally meaningful time intervals.

Splitting & Subsampling:

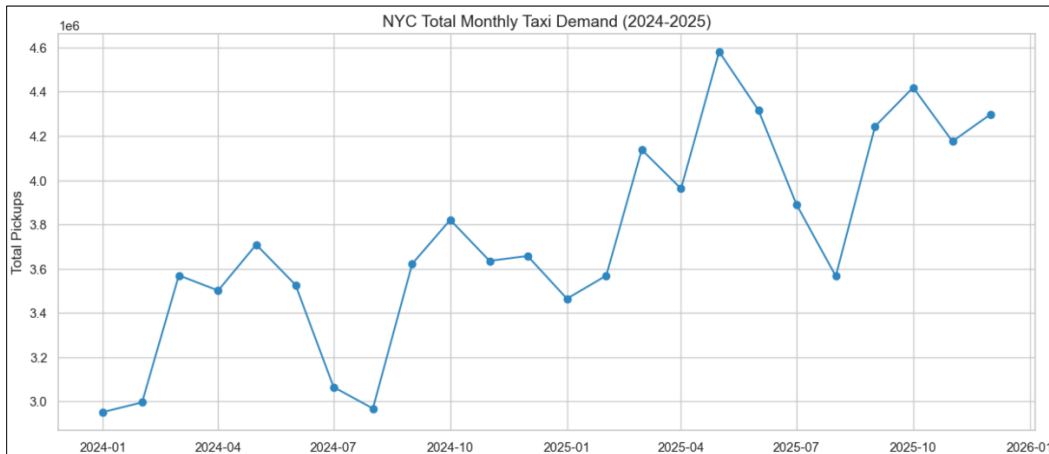
To properly simulate a live production environment where the model encounters unseen future data, we applied a strict splitting strategy:

- **Training Set:** Data spanning from January 2024 to December 2025. This two-year period ensures the model captures robust yearly seasonality and long-term trends.
- **Test / Monitoring Set:** Data from January and February 2026. This serves as our "future" production data to evaluate true generalization and monitor for data drift.
- **Drift Reference Set:** To enable accurate drift detection on the 2026 dataset without exposing the model to its own training data, we extracted a random week from every month across 2024-2025, for a total of 24 weeks. These 24 weeks were completely removed from the training set to serve as an uncontaminated, "In-Distribution" reference baseline for evaluating the 2026 data.
- **Subsampling for Experimentation:** During the iterative model training phase, we utilized randomized subsamples to enable rapid experimentation without compromising the statistical distribution of the dataset.

3. Exploratory Data Analysis:

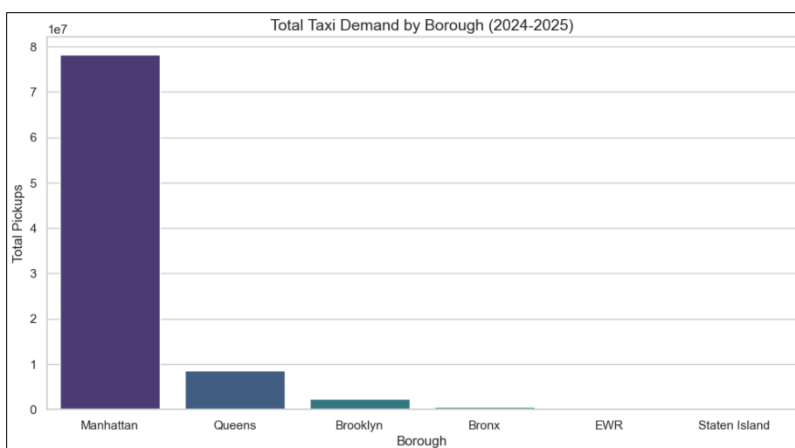
During the Exploratory Data Analysis phase on the training set, we focused on identifying actionable insights that drive both feature creation and business strategy. We handled anomalies, removed invalid location IDs, and uncovered the following critical patterns:

- **Micro and Macro Temporal Seasonality:** The data exhibits clear temporal demand patterns at both the daily and weekly levels. At the micro level, demand consistently peaks during the morning and evening hours, reflecting increased travel activity at the beginning and end of the workday. At the macro level, weekly seasonality reveals markedly different weekend behavior, with late-night demand exceeding early-morning demand in nightlife areas, unlike the patterns observed on weekdays.



This plot illustrates the macro-level seasonality and long-term demand fluctuations across the city during our two-year training period.

- **Geographical Concentration & Disparities:** Demand is highly uneven across locations and exhibits strong geographic concentration. Central Manhattan zones experience consistently high activity levels, while many outer-borough areas see minimal demand during off-peak hours. However, these regions can still experience pronounced localized surges, particularly around airports and major residential or transportation hubs.



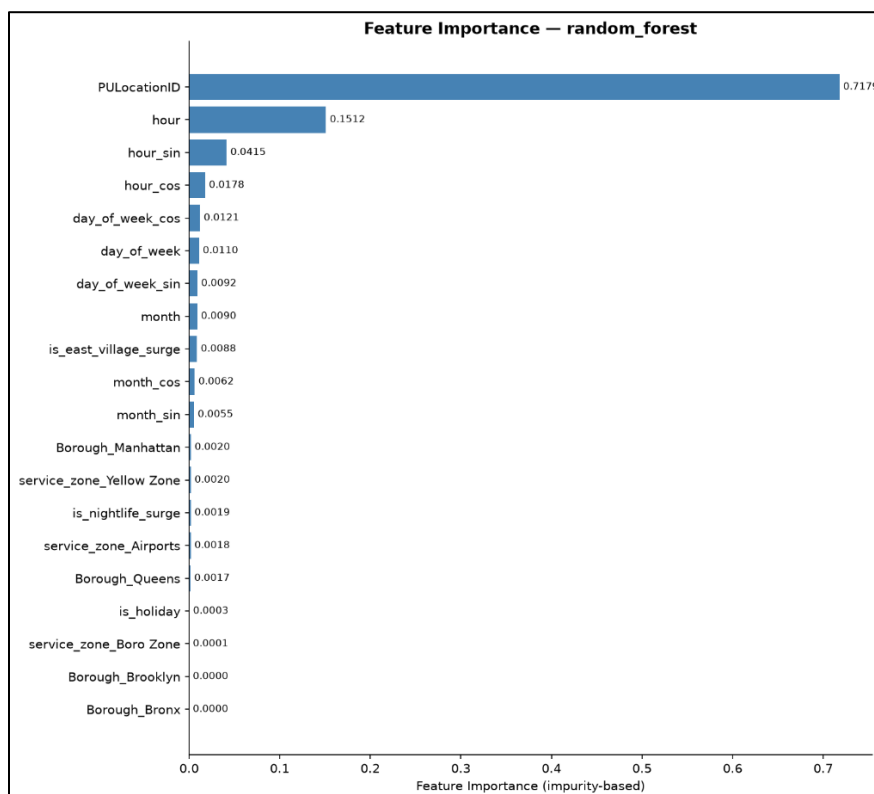
This chart highlights the extreme geographical concentration of the data, demonstrating that the vast majority of taxi pickups occur within Manhattan compared to the outer boroughs.

These insights dictate that Gett's Fleet Operations must adopt a dynamic, predictive allocation strategy. Instead of a uniform city-wide taxi distribution, drivers must be strategically pre-positioned in specific zones ahead of predicted surges. This minimizes "deadheading" and maximizes driver utilization.

4. Feature Engineering:

To improve the model's generalization capabilities, we transitioned from a baseline model (relying solely on structural features: PULocationID, hour, day_of_week, month) to a model enriched with advanced engineered features. We comprehensively mapped our features into these categories:

- **Continuous Temporal Features (Cyclical Encoding):** Time is cyclical. For example, standard linear representation of hours (0-23) misleads the model into assuming the distance between 23:00 and 00:00 is vast. We resolved this by applying Trigonometric Encoding (Sine/Cosine) to create the features hour_sin, hour_cos, month_sin, month_cos, day_of_week_cos and day_of_week_sin.
- **Geographical Enrichment (Categorical Encoding):** We merged our data with the external Taxi Zone Lookup table. Instead of relying solely on an arbitrary numerical identifier (PULocationID), we engineered two new categorical features: Borough and service_zone. These were numerically encoded to provide the model with broader geographic context.
- **Location-Specific Demand Patterns (Target Encoding):** Rather than using the raw PULocationID as an arbitrary identifier, we replaced it with the average number of pickups per 30-minute interval for each location, enabling the model to capture historical demand patterns associated with different pickup zones.
- **Event & Nightlife Anomalies (Custom Logic):** To help the model understand anomalous human behavior, we engineered highly specific boolean flags based on domain knowledge:
 - a) **Is_holiday:** Flags major U.S. holidays, which often exhibit atypical travel demand patterns.
 - b) **Is_nightlife_surge:** Flags specific general nightlife hotspots during prime late - night hours (22:00–04:00) on weekends and New Year's Day.
 - c) **Is_east_village_surge:** A dedicated flag for the East Village (PULocationID=79) to capture its highly unique and extreme weekend surge behavior.
- **Feature Scaling:** To ensure stability across algorithms, all numerical and cyclic features were normalized using a Standard - Scaler as the final step in the feature pipeline.



This plot illustrates the relative importance of features during initial model evaluation, highlighting which inputs contribute most to the model's predictive performance.

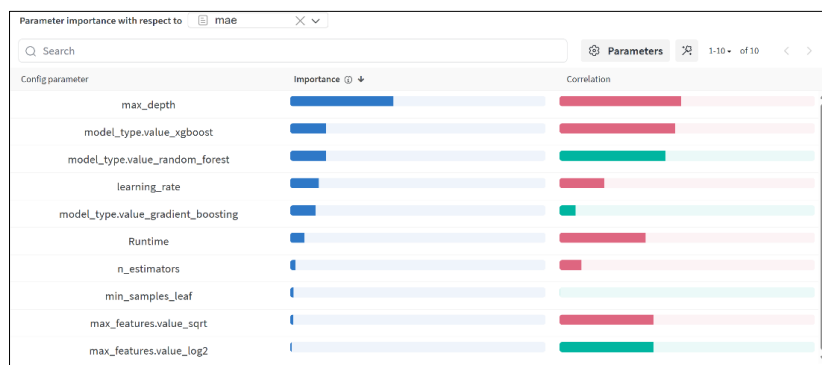
5. Model Selection & Hyperparameter Tuning:

All experiments were rigorously tracked using the **Weights & Biases** platform. We evaluated multiple machine learning families (Linear Regression, Random Forest, XGBoost, Ridge). The primary evaluation metric chosen was **MAE**, as it provides an intuitive "number of missing/excess taxis" metric for business operations.

The Tuning Process:

- **Phase 1 (Random Search):** We ran a broad W&B Sweep on a data subsample to identify the *Model Family* that performed best out-of-the-box with our engineered features.
- **Phase 2 (Grid Search):** Once the Champion Model family was identified, we performed a focused Grid Search on specific hyperparameters to extract maximum performance.

Result: Training the tuned champion model on the engineered feature set yielded a significant MAE improvement compared to the structural baseline model.



This dashboard view demonstrates our tuning process and tracks the significant reduction in Mean Absolute Error (MAE) achieved by our optimized champion model compared to the structural baseline.

6. Model Improvement Loops (Automated CI/CD, Error Analysis & Drift Management):

To ensure the model remains effective in a dynamic production environment, we implemented a comprehensive model improvement loop that extends beyond initial training and evaluation. The system continuously analyzes prediction errors to identify systematic weaknesses, monitors incoming data for feature and concept drift, and automatically applies mitigation strategies when significant distribution shifts are detected. By combining automated monitoring, error analysis, and drift management, the pipeline transforms model maintenance from a manual process into a continuous feedback cycle that supports long-term reliability and performance.

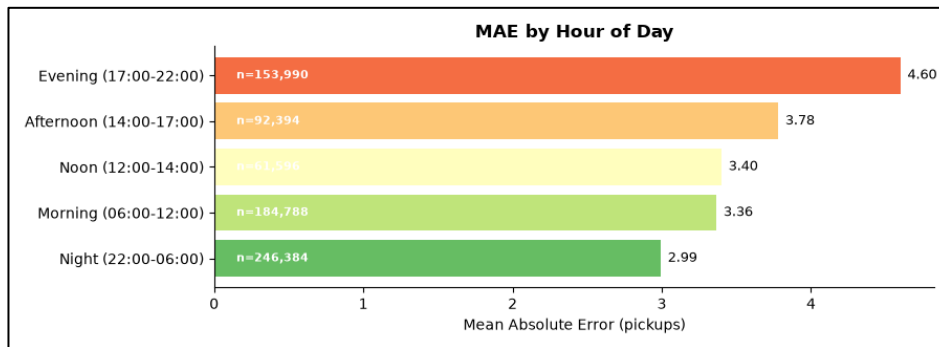
The Automated Pipeline Loop:

Our system is orchestrated to execute the full end-to-end ML pipeline automatically:

- **Triggers:** The pipeline is automatically triggered whenever changes are pushed to the main branch that affect the source code (src/), orchestration script (pipeline.py), dependencies (requirements.txt), or raw data (data/raw/).
- **Continuous Logging:** During each pipeline run, the system retrains and fine-tunes candidate models, evaluates their performance, and performs comprehensive error analysis. Residual metrics, error distributions, hyperparameter tuning results, and evaluation visualizations are automatically logged to Weights & Biases, enabling continuous tracking and improvement of model performance.
- **Continuous Monitoring:** Following the error analysis, the pipeline automatically triggers our Drift Detection protocols to evaluate the health of incoming live data.

Example of Error Analysis - Error by Time of Day:

By analyzing the W&B logged errors, we identified specific temporal segments where the model struggles most. Instead of looking at arbitrary individual hours, we aggregated the data into distinct parts of the day. The analysis reveals that the highest Mean Absolute Error (MAE) occurs during the Evening segment, correlating with the volatile rush hour and post-work period, while the Night segment remains the most predictable.

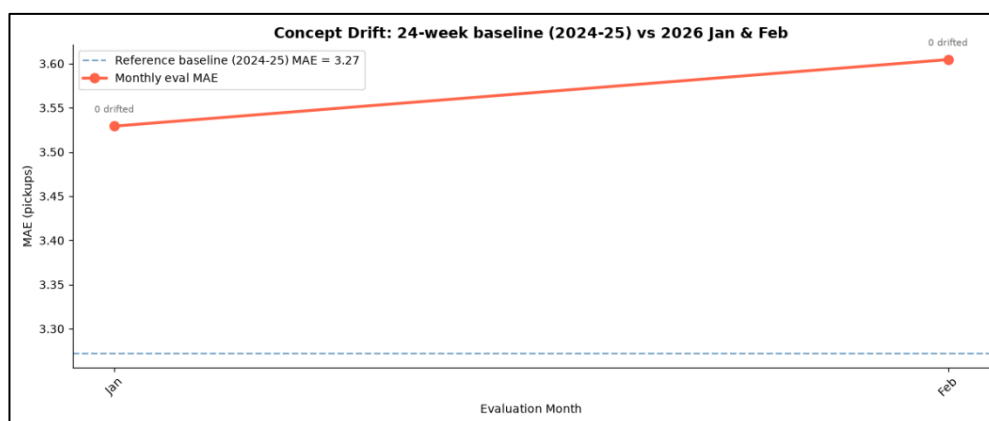


This error analysis chart demonstrates that our model's predictive difficulty peaks during the Evening segment indicating highly volatile demand during post-work hours, whereas Night-time demand remains relatively stable and predictable.

Drift Detection and Mitigation:

Treating the Jan-Feb 2026 data as live production data, the automated pipeline compares it against the isolated 24-week Reference Set from 2024-2025.

- Drift Detection:** The pipeline monitors feature, target, and concept drift. Feature drift is detected using PSI, KS tests, and Evidently AI reports. Target drift is assessed by comparing demand distributions between incoming 2026 data and reference set. Concept drift is evaluated by comparing model performance on 2026 data against the reference set.
- Automated Mitigation:** When significant drift is detected for a feature, the pipeline automatically removes the drifted feature and retrains the model from scratch. By excluding unstable features, the model becomes less sensitive to distribution shifts and can maintain more reliable predictive performance on new data.



This curve indicates an approximately 8-10% increase in MAE when evaluated the model on the unseen 2026 production data (red line) relative to the historical 24-week baseline MAE of 3.27 (blue dashed line).

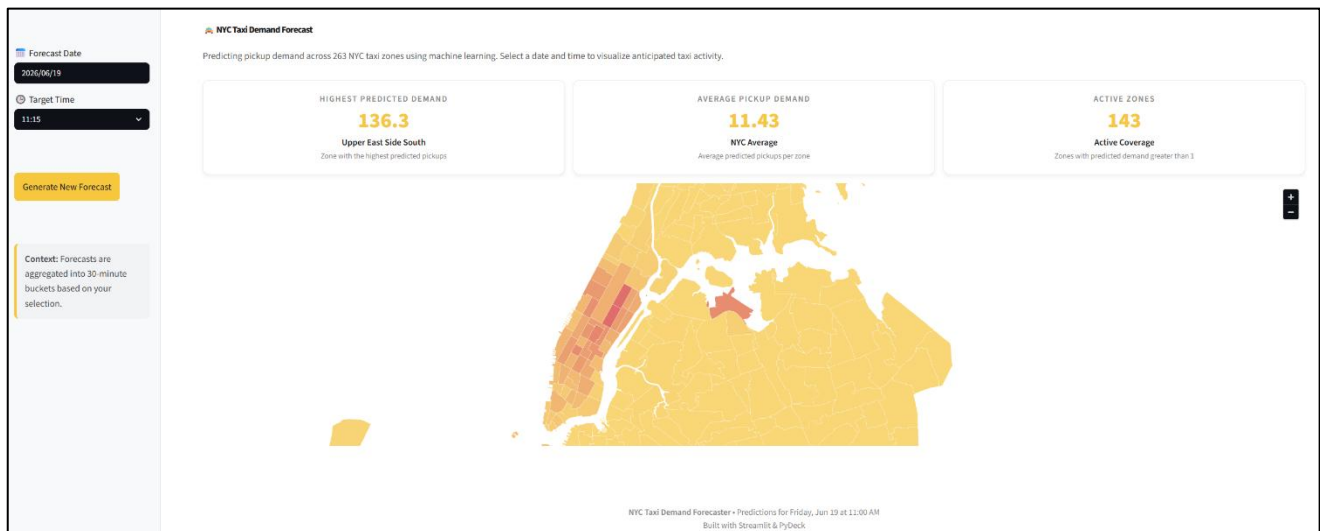
7. User Interface:

While our automated pipeline ensures backend stability, technical reports alone are insufficient for fleet managers making real-time decisions on the ground. To bridge the gap between complex Machine Learning models and actionable operational decisions, we developed a production - grade, interactive web dashboard using Streamlit and PyDeck.

The Forecasting Dashboard:

The dashboard serves as the primary tool for Gett's Fleet Operations. It connects directly to our tuned champion model and allows users to:

- **Query Future Demand:** Users can input a specific future date and time (up to 364 days in the future) via the sidebar, which the system automatically translates into our operational 30-minute forecasting buckets.
- **Analyze High-Level Metrics:** The dashboard instantly calculates and displays critical operational metrics, including the specific zone with the Highest Predicted Demand, the NYC Average Pickup Demand, and the total number of Active Coverage Zones.
- **Visualize the Geo-Spatial Heatmap:** Using PyDeck, the dashboard renders an interactive heatmap of all 263 NYCTaxi Zones. Zones are color-coded based on predicted demand, enabling rapid identification of high-demand areas, while interactive tooltips display exact pickup forecasts for each zone.



Conclusion:

This project presents an end-to-end machine learning pipeline for forecasting New York City taxi demand, covering data preprocessing, feature engineering, model development, hyperparameter tuning, error analysis, and drift monitoring. By integrating automated drift detection and mitigation, the pipeline provides a practical framework for maintaining model performance as data distributions evolve over time.